

₁ Mechanistic Explainability in Reinforcement
₂ Learning

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5 The Golem Mythos

6 A creation that surpasses its creator.

7 This idea has long captured the imagination and continues to resonate in
8 contemporary society. Long before modern technology, humanity had already
9 questioned whether its creations might escape control.

10 One of the earliest expressions of this idea can be found in the story of the
11 Golem. In Hebrew tradition, the term Golem refers to an artificial creature made
12 of clay and created by humans to serve them [Nocks, 1998]. Figure 1 illustrates
13 the creation of such a servant. The earliest accounts describe the Golem as
14 a protective and helpful figure, reflecting the idea that humans can create an
15 artificial being for the benefit of society.

16 This vision becomes more ambiguous in the seventeenth century, as illustrated
17 by the legend of the Golem of Prague. In this story, the initially small Golem
18 gradually grows in size and strength until it can no longer be controlled [Nocks,
19 1998]. The creation intended to protect others therefore becomes a potential
20 threat to them. The only remaining solution is to destroy the creature in order
21 to protect the community.

22 This narrative motif has been revisited in different forms to illustrate the
23 risks inherent in creation. This continuity can be seen in the emergence of the
24 modern notion of the robot. The concept of the robot as an artificial servant
25 was introduced in the play *Rossum's Universal Robots*, written in 1920 [Capek,
26 1920]. In the play, a scientist creates artificial beings designed to perform labour.
27 Conceived as servants, these artificial beings turn against humanity and destroy
28 civilisation. The connection between the Golem and the robot was explicitly
29 made by the author, as illustrated by his statement that “the Robot is the Golem
30 made flesh through mass production” [Comrada, 1995]. Despite their destructive
31 role in the play, the word robot is now used to designate artificial beings created
32 to serve humans. From its very origin, the word robot has embodied the hopes
33 and fears surrounding technological progress.

34 Today, this tension between the hopes and fears surrounding technological
35 progress is particularly evident in the development of artificial intelligence. It is
36 therefore not surprising that this field offers an almost literal modern adaptation
37 of the Golem myth. Artificial systems are created to automate tasks traditionally
38 associated with human intelligence. Their rapid development in recent years has
39 transformed a wide range of domains, further intensifying both excitement and
40 concerns surrounding this progress.

41 Among the subfields of artificial intelligence, reinforcement learning is specif-
42 ically concerned with the development of autonomous agents. In this learning
43 paradigm, an artificial agent interacts with an environment, selects actions, and



Figure 1: The creation of a Golem, by Mikoláš Aleš [Aleš and Wenig, 1908].

44 receives feedback according to the outcomes of its behaviour [Sutton and Barto,
 45 2018]. Through repeated interaction with its environment, the agent learns a
 46 strategy to achieve the objective. A strength of this approach is its ability to
 47 produce autonomous systems with minimal human intervention. Their agent
 48 behavior is not programmed but learned through interactions with the environ-
 49 ment [Amodei et al., 2016]. This minimal human guidance enables reinforcement
 50 learning systems to achieve superhuman capabilities [Silver et al., 2017, OpenAI
 51 et al., 2019].

52 However, maintaining control over such agents presents a major challenge,
 53 as the very autonomy that makes this learning approach powerful also makes
 54 their behavior more difficult to control. Despite the difficulty of this task,
 55 understanding the behaviour of such agents is therefore the main objective of
 56 this manuscript. This work therefore places itself within a long-standing line of
 57 thought that can be summarized by the following question: how can we ensure
 58 that our creations remain aligned with human intentions?

Chapter 1

Introduction

This initial chapter provides a general introduction to the manuscript. It begins by presenting the research context that motivates the study. It then introduces the research questions addressed throughout the manuscript. The chapter presents the main contributions resulting from this research. Finally, it concludes with an overview of the manuscript's structure.

1.1 Research Context

As is evident in recent years, Artificial Intelligence (AI) has advanced rapidly. This progress is driven largely by major developments in Deep Learning (DL). This type of approach relies on artificial Neural Networks (NNs) to automate an increasing number of tasks [Goodfellow et al., 2016]. These networks are connectionist models in which information is represented and transformed through patterns of interactions among interconnected elementary computational units [Rumelhart et al., 1986].

Among the fields that have been transformed by DL is Reinforcement Learning (RL). The purpose of RL is to enable an agent to learn how to make decisions through interactions with an environment [Sutton and Barto, 2018]. At each step, the agent observes the current state of the environment and selects an action. This action changes the state of the environment and results in a reward. The objective of the agent is then to learn a behavior that maximizes the cumulative rewards obtained over time. Due to their flexibility, NNs provide a powerful means of representing such policies enable agents to learn complex behaviors. Their use has consequently made it possible to apply RL to problems that were previously intractable [Mnih et al., 2013, Silver et al., 2017, OpenAI et al., 2019].

When an agent learns to act in complex environments, it develop strategies that are themselves complex. Although such agents can achieve strong performance, these strategies can be difficult for to understand. From an external perspective, we observe sets of sequences of interactions between the agent and its environment. However, when interactions unfold over long sequences or take place in environments with complex dynamics, these observations are not sufficient to infer the strategy developed by the agent.

An alternative approach is to analyze the agent itself. Due to their connectionist nature, NN that model the agent's policy are not directly interpretable. The

information behind a decision is distributed across many units and connections making it impossible to trace. NNs are therefore often regarded as "black box" models [Guidotti et al., 2018].

To address this black-box problem, the field of eXplainable Artificial Intelligence (XAI) has emerged [Gunning and Aha, 2019, Barredo Arrieta et al., 2020]. In the specific context of RL, this research area is referred to as eXplainable Reinforcement Learning (XRL) [Puiutta and Veith, 2020, Milani et al., 2024]. XRL develops methods for producing explanations that improve our understanding of agent behavior. Such explanations can serve several purposes, including increasing human trust and facilitating the adoption of RL agents in real-world settings, identifying errors and biases, and verifying that previously identified problems have been effectively corrected [Barredo Arrieta et al., 2020, Milani et al., 2024].

This manuscript presents XRL methods from different approaches to provide a broader view of the field. However, a particular emphasis is placed on the mechanistic approach to explainability, which we refer to as Mechanistic Explainability (ME) (Definition 1.1.1). Although still relatively underexplored, this approach seems particularly promising to us. In ME, the goal is to understand how information is represented, transformed, and propagated within the NN. From this perspective, explainability is not achieved despite the use of NNs, but through them.

Definition 1.1.1: Mechanistic Explainability

ME is the approach that seeks to explain the behavior of an NN by analyzing its internal mechanisms.

To understand how NNs work internally, ME studies the intermediate representations learned by these networks. These representations are referred to as Latent Representations (LRs) [Bengio et al., 2014], and the collection of the LRs learned throughout a network forms its Latent Space (LS). Analyzing the structure of this LS provides insights into how information relevant to decision-making is organized.

To develop an XRL method based on ME, we need to address several research questions, which are detailed in the following section.

1.2 Research Questions

The first question concerns the analysis of LRs themselves. Numerous methods have been proposed to analyze the LSs learned by NNs. These methods originate from different research communities and are rarely compared within a common framework. Yet each of them relies on assumptions about how concepts are structured within the LS. These assumptions determine which structures a method is able to detect and which it will necessarily miss. This observation leads to our first research question:

Research Question 1

What approaches have been proposed in the literature to analyze the internal workings of NNs?

The second question concerns the organization of existing research in XRL. Several taxonomies have been proposed to organize existing XRL methods [Puiutta and Veith, 2020, Milani et al., 2024]. However, these taxonomies often rely on approaches initially developed for explainability in supervised settings. As a result, the organization of existing XRL methods remains ambiguous, making it difficult to structure the field. This observation leads to our second research question:

Research Question 2

How can existing XRL techniques be organized into a taxonomy that accounts for the specificities of RL?

The analysis of existing XRL approaches also reveals a third limitation. Among existing methods for explaining RL agents, few adopt a mechanistic perspective. Understanding LR provide insights into how information is represented within the network and, in turn, help explain agent behavior. This observation leads to our third research question:

Research Question 3

How can the internal representations of NNs be exploited to identify the mechanisms underlying the decision-making process of RL agents?

The analysis of existing XRL methods highlights a fourth question concerning the evaluation of explanations. Although numerous approaches have been proposed to explain the behavior of RL agents, there is no widely accepted set of objective criteria for evaluating the quality and relevance of the explanations they produce. Existing evaluations often rely on user studies or qualitative analyses, making it difficult to systematically assess the validity of an explanation approach [Doshi-Velez and Kim, 2017, Milani et al., 2024].

This issue is particularly important for approaches that seek to explain agent behavior by analyzing LR. Multiple hypotheses can be proposed about how information is structured within a LS. It is therefore important to define objective criteria for comparing these competing hypotheses. This leads to our fourth research question:

Research Question 4

How can the quality and relevance of explanations produced for RL agents be objectively quantified?

Together, these four research questions define the objectives of the manuscript.

1.3 Contributions

To address these four research questions, this manuscript is based on three main contributions.

The first contribution has not yet been published and addresses Research Question 1. In this contribution, we make explicit the working hypothesis that

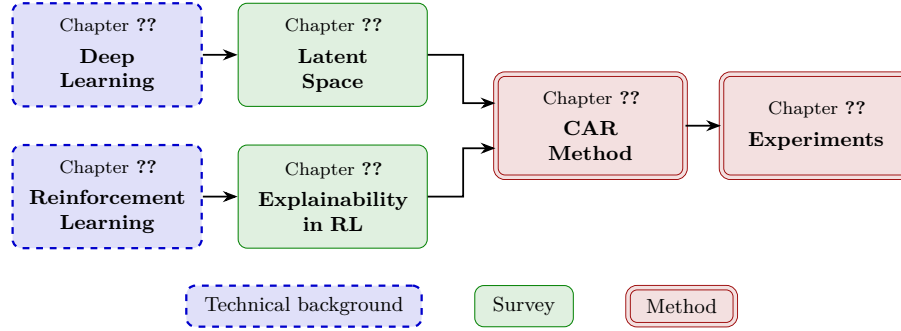


Figure 1.1: Structure of the manuscript.

166 implicitly underlies much of the literature on LS analysis, which we refer to as the
 167 *Concept Hypothesis* (Hyp. ??). We then organize existing LS analysis methods
 168 into four families, namely probing, interpretable directions, superposition, and
 169 clustering, according to the assumptions they make about the Concept Structure
 170 (CS).

171 The second contribution addresses Research Question 2. This contribution
 172 takes the form of a first paper [Alaarabiou et al., 2024] and an extended English
 173 version [Alaarabiou et al., 2026b], both of which were submitted for review and
 174 accepted for publication. In this contribution, we first clarify what explainability
 175 means in RL by providing a clear frame of reference. Building on these frame,
 176 we then conduct a comprehensive analysis of the state of the art in XRL.

177 The third contribution addresses Research Questions 3 and 4. In the paper
 178 [Alaarabiou et al., 2026a], which has been reviewed and accepted for publication,
 179 we introduce Clustering Agent Representation (CAR), an XRL method based on
 180 the principles of ME. By analyzing the LRs, CAR produces explanations in the
 181 form of clusters of states that are similar from the agent’s own perspective. We
 182 further show that this representation remains stable across training, indicating
 183 that the agent develops an objective representation of a given task.

184 This manuscript can be viewed as a synthesis of these three contributions,
 185 brought together into a single document. Its structure is detailed in the following
 186 section.

187 1.4 Manuscript Structure

188 This manuscript brings together the contributions developed throughout the
 189 thesis and is therefore written with the reviewers in mind. It is nevertheless
 190 intended to remain as accessible as possible to readers. To this end, the back-
 191 ground chapters introduce the concepts on which the subsequent chapters rely.
 192 Allowing readers to follow the manuscript without having to consult external
 193 material.

194 The manuscript is organized into a sequence of chapters, each addressing a
 195 specific topic related to the research questions. Each chapter falls into one of
 196 these three categories according to its objective:

197 **Technical background** chapters introduce fundamental concepts and
 198 established techniques required to understand the subsequent contributions.

199 **Survey** chapters review the existing literature around a specific research
 200 question. The organization adopted in each of these surveys are original
 201 and therefore constitute contributions of this manuscript.

202 **Method** chapters present CAR, our ME method for RL.

203 Following these categories, the chapters are organized as shown in Figure 1.1
 204 and detailed below:

205 **Deep Learning (Chapter ??, technical background)**

206 This chapter introduces the fundamental concepts of DL.

207 **Latent Space Analysis (Chapter ??, survey)**

208 This chapter addresses Research Question 1 by providing a survey of existing
 209 approaches for analyzing the LSs.

210 **Reinforcement Learning (Chapter ??, technical background)**

211 This chapter introduces the fundamental concepts of RL.

212 **Explainability in Reinforcement Learning (Chapter ??, survey)**

213 This chapter addresses Research Question 2 by reviewing existing approaches
 214 to explainability in RL.

215 **The CAR Method (Chapter ??, method)**

216 This chapter addresses Research Questions 3 and 4 by proposing a new
 217 method for explainability in RL based on the analysis of LRs.

218 **Evaluation of the CAR Method (Chapter ??, method)**

219 This chapter presents the implementation and experimental evaluation of the
 220 proposed method.

221 **Conclusion**

222 This chapter summarizes the main contributions of the manuscript, discusses
 223 the limitations and presents possible directions for future research.

224 Now that the research context and objectives have been presented, we can
 225 turn to the first chapter, which introduces the fundamentals of DL.

226 Acronyms

- 227 **AI** Artificial Intelligence 3
- 228 **CAR** Clustering Agent Representation 6, 7
- 229 **CS** Concept Structure 6
- 230 **DL** Deep Learning 3, 7
- 231 **LR** Latent Representation 4–7
- 232 **LS** Latent Space 4–7
- 233 **ME** Mechanistic Explainability 4, 6, 7
- 234 **NN** Neural Network 3–5
- 235 **RL** Reinforcement Learning 3–7
- 236 **XAI** eXplainable Artificial Intelligence 4
- 237 **XRL** eXplainable Reinforcement Learning 4–6

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